Turbine: Building Model 2+1 Web Applications

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Turbine: Building Model 2+1 Web Applications

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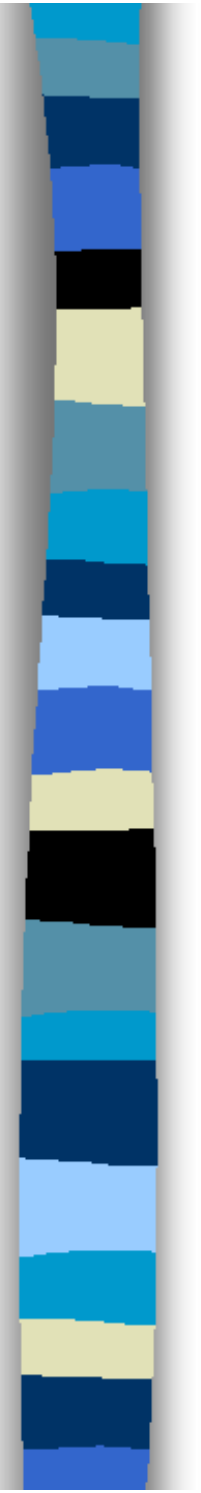
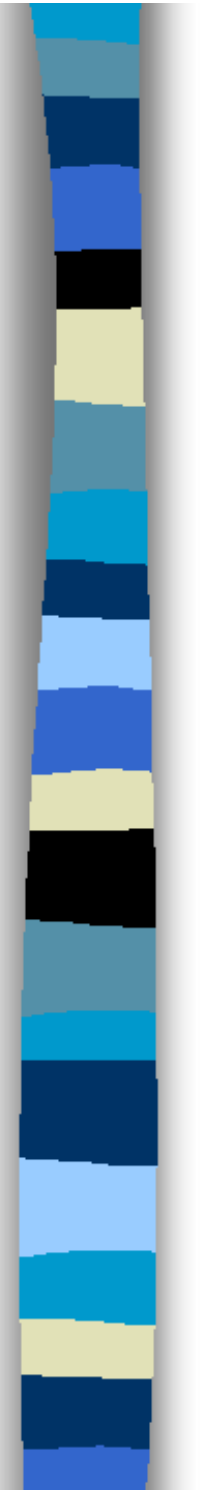


Outline

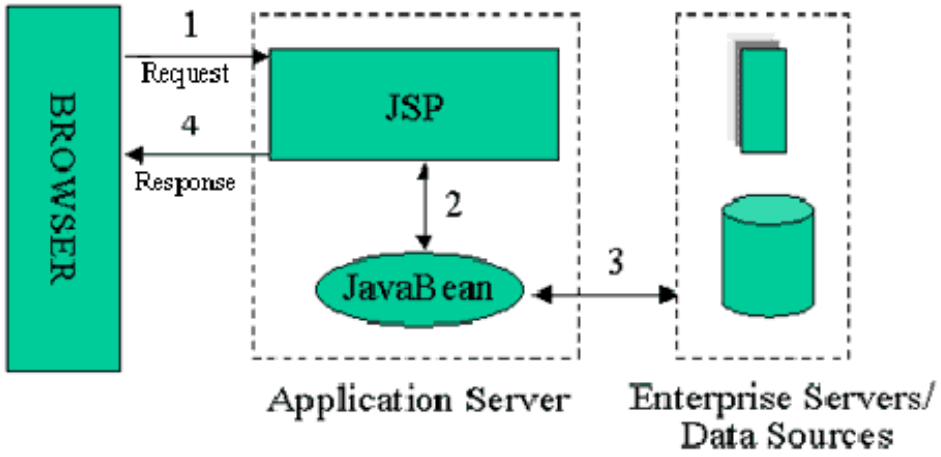
- What is Model 1?
- What is MVC?
- What is Model 2?
- What is Turbine (Model 2+1)?
- How do I build applications with Turbine? (TDK)

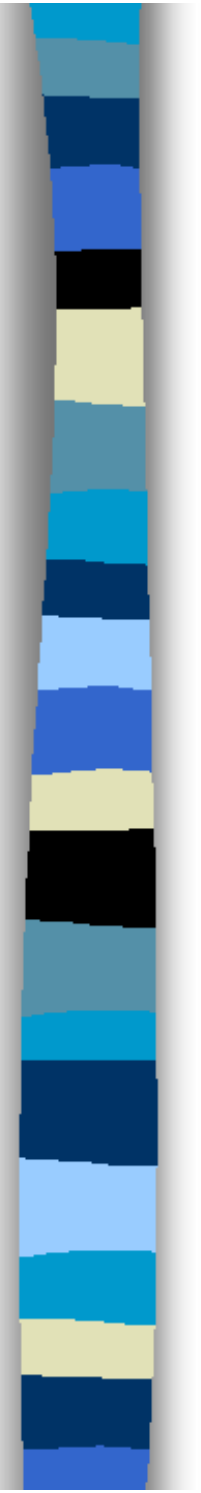


What is Model 1?

- Model 1 is the current method of creating sites using JSP/Servlets. A linear and non-expandable approach.
- Flow works like this:
  - Request comes in
  - JSP page executes Beans to do processing
  - Result is returned

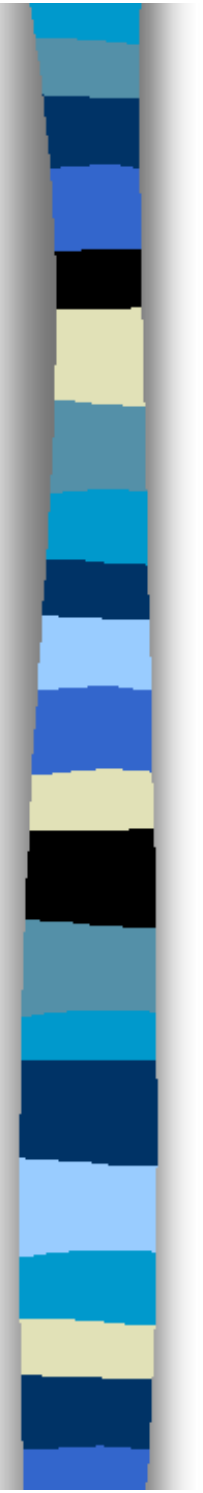
Picture of Model 1





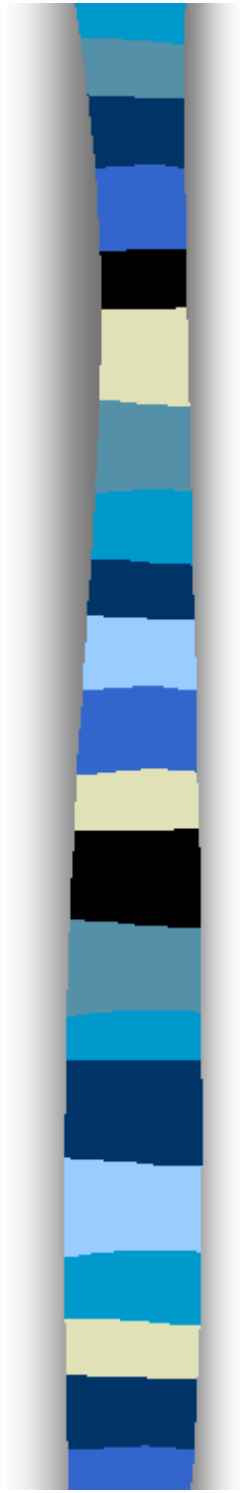
What is MVC?

- Model-View-Controller
- Abstraction of each layer of development
- Model - Application Logic
- View - Presentation Logic
- Controller - Business Logic



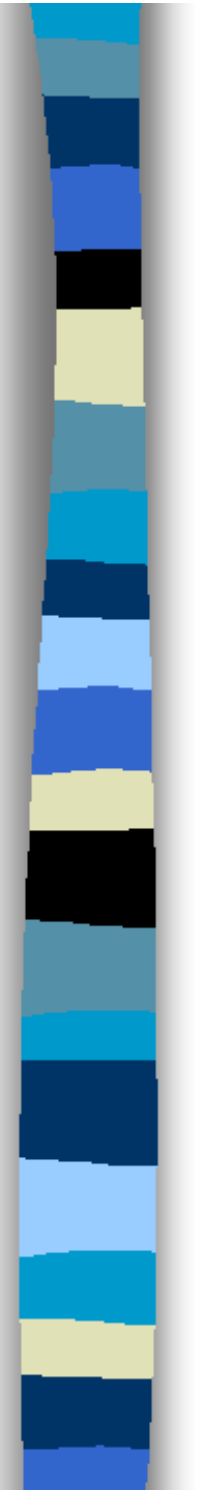
Model - Core of the Application

- Application Logic
  - database accesses
  - data crunching/munging
  - Separation of UI from application
- Not dependent on input or output formats



View - Presentation Layer

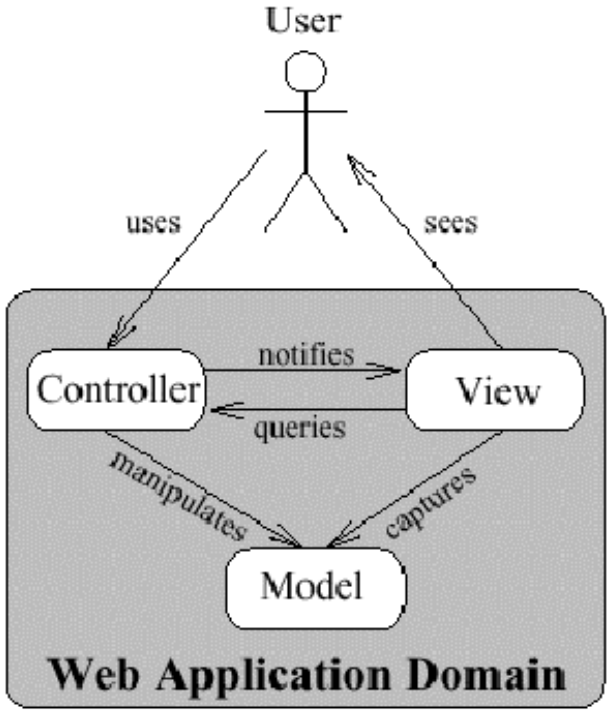
- Responsible for displaying the combined work of the Controller and Model
- UI portion of the page
- Formats data to be presentable to the end user



Controller - Input Mechanism

- Listens to what the User requests
- Supplies and communicates information to the Model and the View

A Picture of MVC



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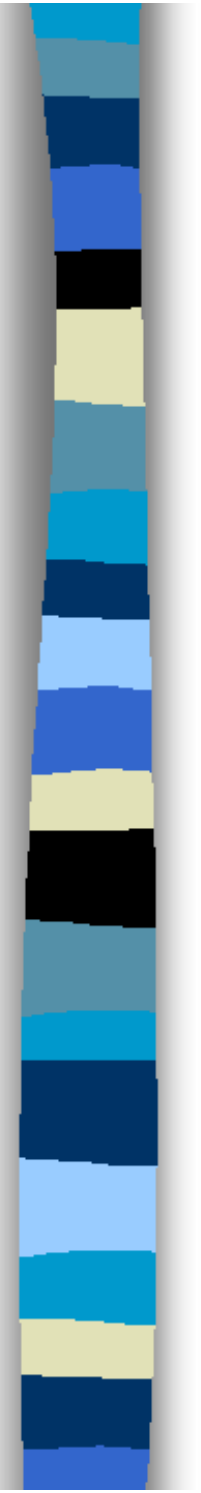
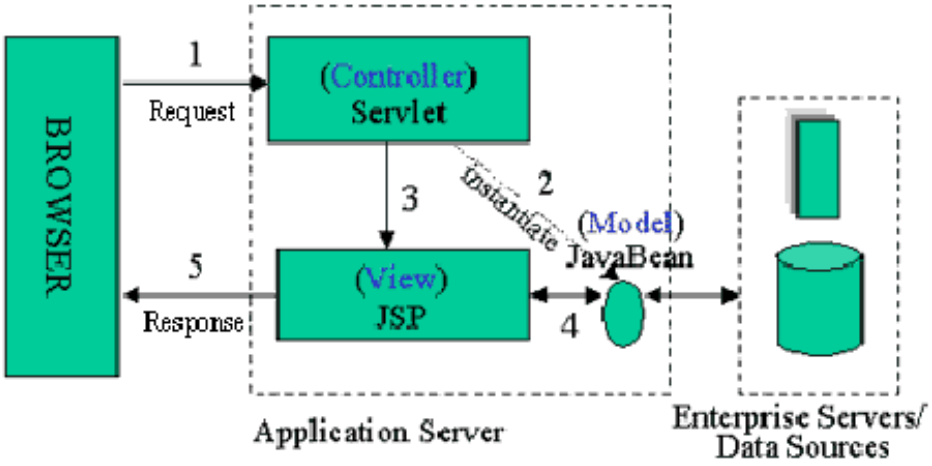


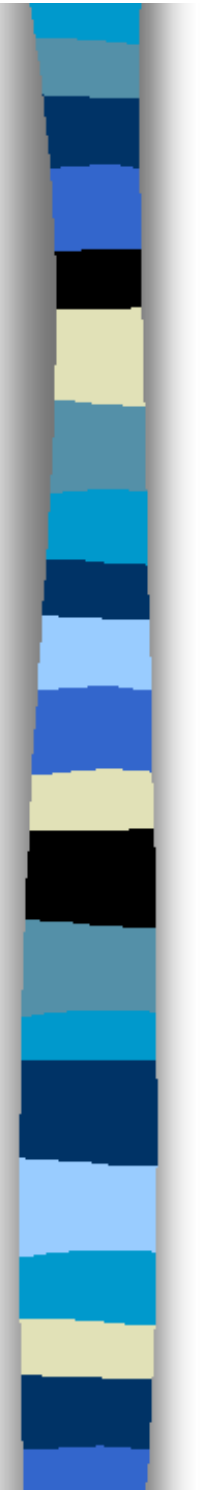
What is Model 2?

- It is a server side implementation of MVC using Java Server Pages (JSP)
- Flow works like this:
  - Request comes in
  - Servlet (Controller) is executed to build a Context (JavaBean)
  - Context is passed to JSP template to provide the View
  - Result is returned

Picture of Model 2

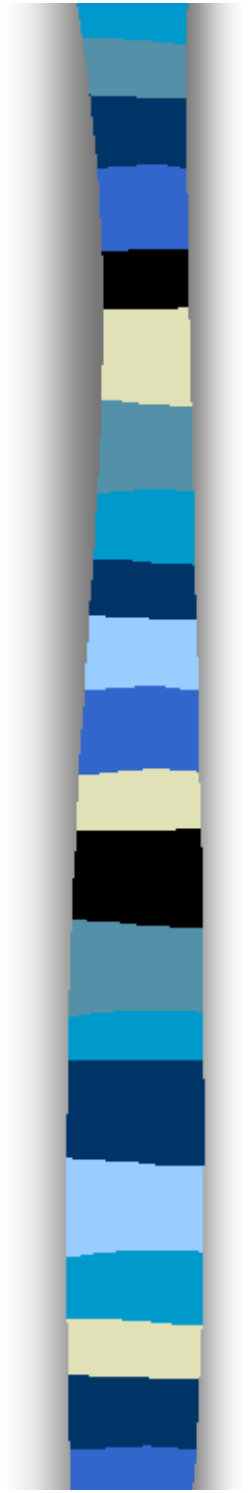


Picture of Model 2



Problems with Model 2

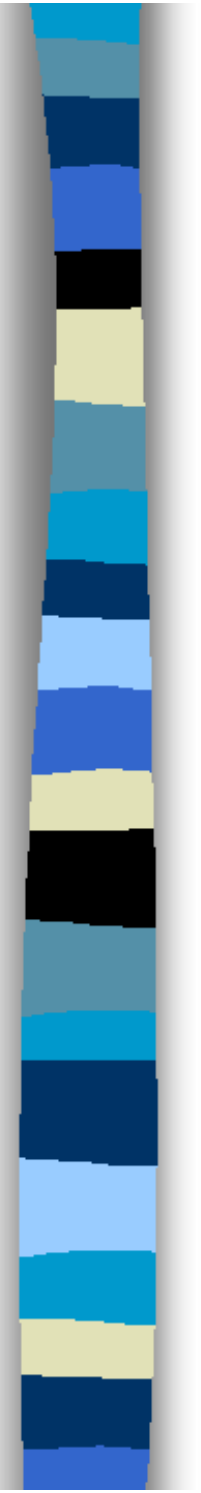
- Does not address the issue of providing a consistent View across the application
- Does not mention a framework for re-usable code.
- Does not address issue of tying the Controller to the View
- Does not address issue of receiving input from the View
- Needs to be more web application friendly



A JSP solution to Model 2 Problems

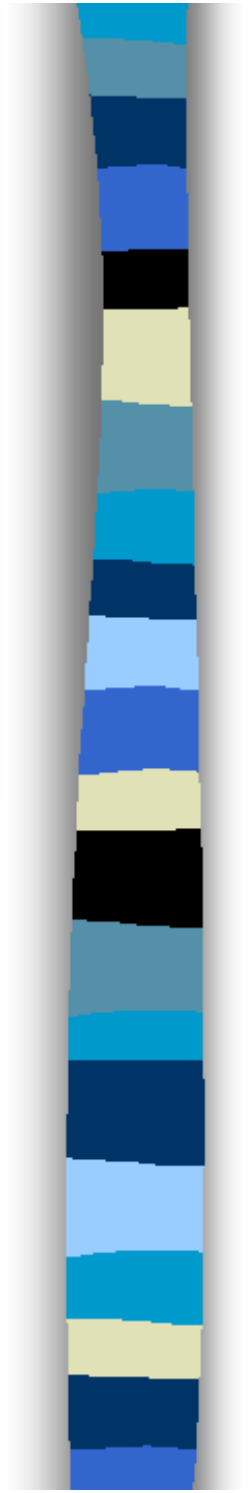
- For those of you who refuse to use anything other than JSP, the Struts project is the right way to go for now.

- Http://jakarta.apache.org/struts/



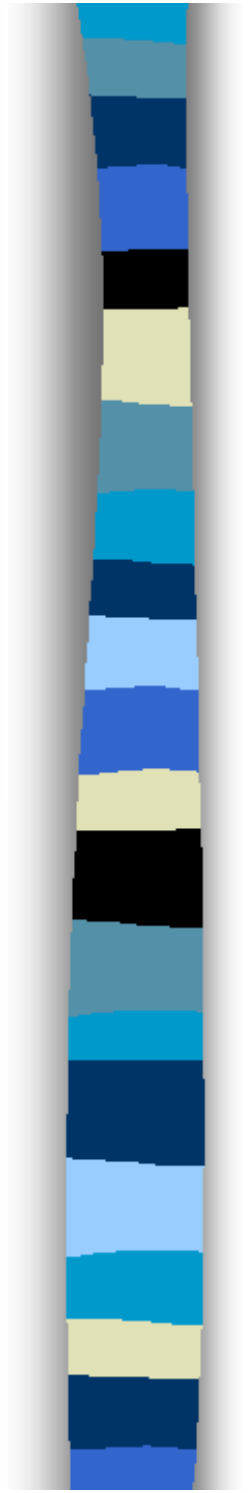
What is Turbine?

- Turbine is a servlet based framework that allows experienced Java developers to quickly build secure web applications.
- A web based application is an application where users use their favorite web browser in order to access secure business logic.



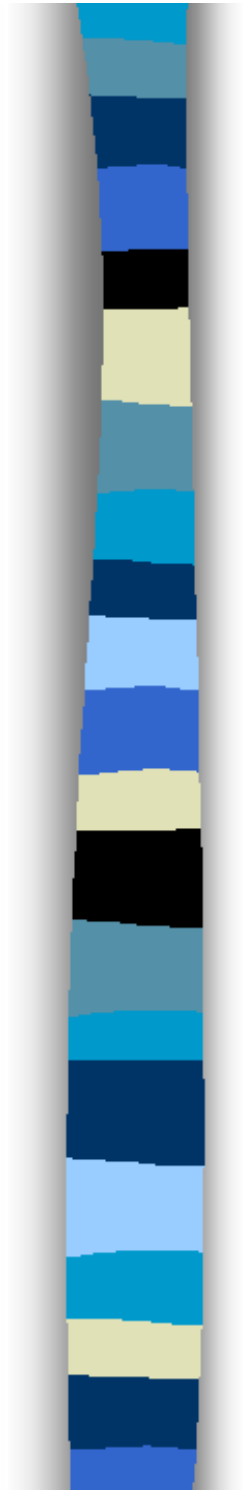
Problems Turbine Solves

- Turbine attempts to not only provide a framework for development, but it also tries to provide all of the code that you would normally end up developing yourself anyway.
- Turbine is trying to be the wheel. Anything else is re-inventing that wheel.
- Turbine has a Module architecture for developing web applications.



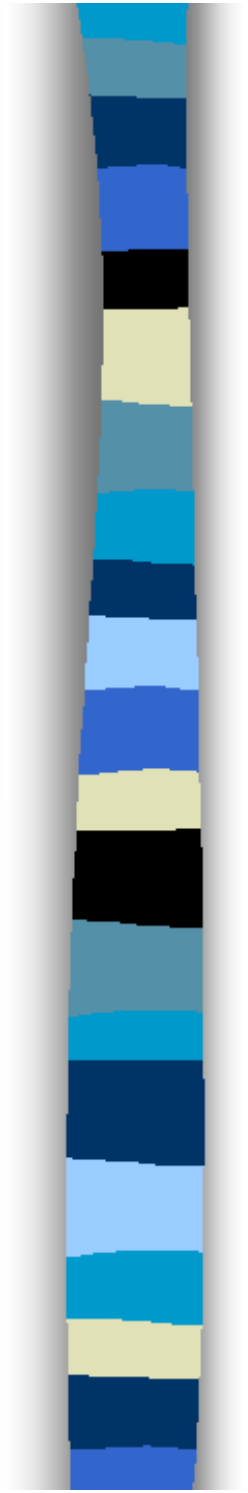
Turbine Features 1

- Integration with template systems: WebMacro, Velocity, Cocoon, FreeMarker, JSP, ECS
- Utility code for working with WM/Velocity, such as a SelectorBox class for building <select> boxes
- Single Entry Point Servlet model for optimal security and control
- Singleton based Database Connection Pool (JDBC) with built in support for all of the major databases
- ParameterParsing for GET/POST/PATH_INFO
- Event based Action handling!



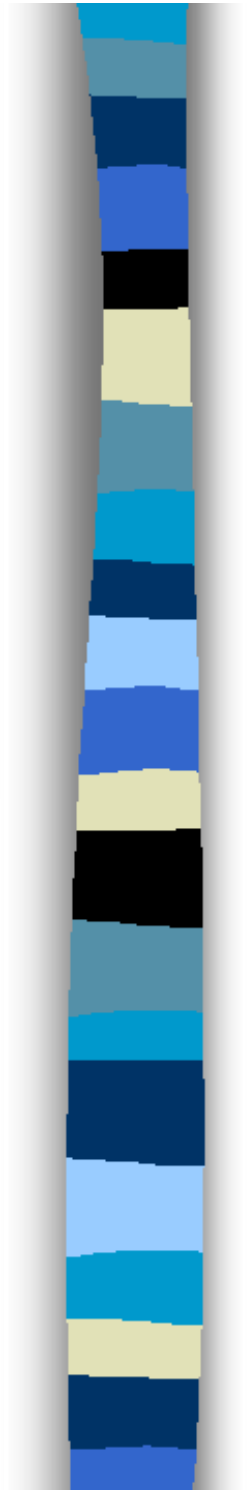
Turbine Features 2

- Strict MVC guidelines implemented through many interfaces and abstract classes as well as through the template systems.
- Integration with 5 Object-Relational tools: DatabaseMap/Peers, Village, Town, OPaL, Castor
- IDBroker that abstracts the autoinsert/sequence usage from the database
- DatabaseMap generation tool that will read a schema and generate the OR Java classes for you
- Failsafe Job based scheduler system (ie: a Java based cron)



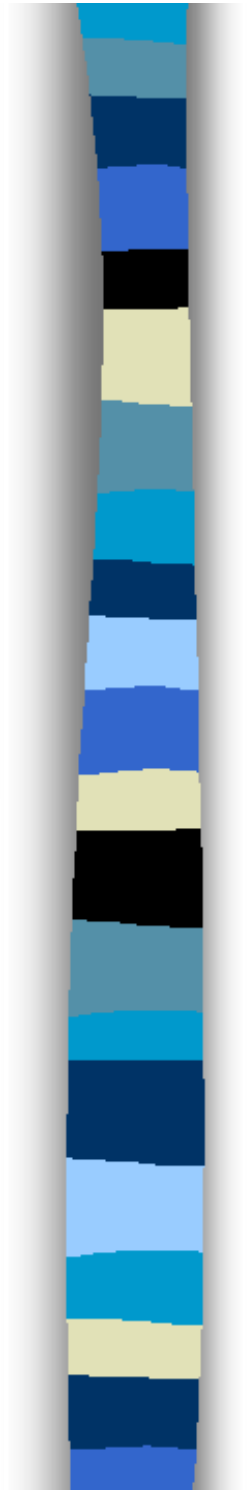
Turbine Features 3

- GlobalCache (Singleton based system for caching data across servlets and requests)
- DateSelector (utility for building the HTML for popup date menus)
- File Upload API
- Generic Services API for creating Singletons
- Castor Service Integration
- XML-RPC Service Integration
- Localization Services API
- JNDI Services API



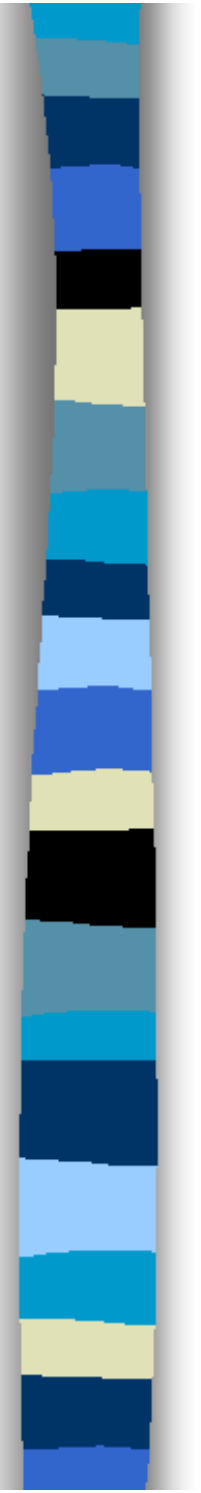
Turbine Features 4

- Visitor/Member API for managing users
- Temporary and Permanent storage/management for users session data
- ACL (Access Control List) based security system that uses Roles and Permissions (and the database schemas)
- BrowserDetector class for determining which browser someone is using
- GenerateUniqueId class for getting a unique number (ie: for sessions)
- Logging to a file facility.



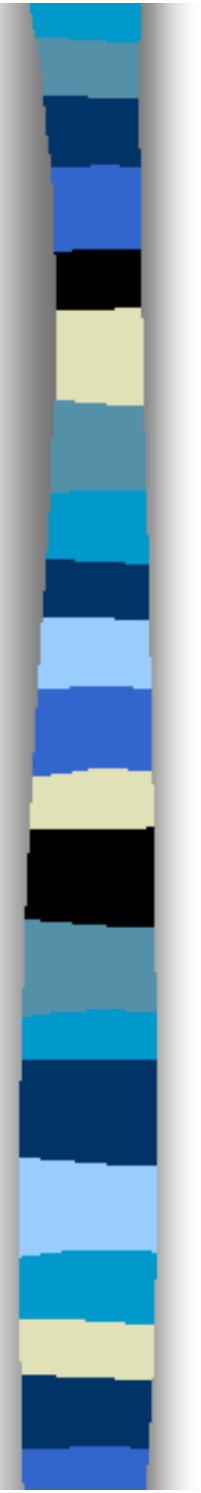
Turbine Features 5

- Property file reader tools and a TurbineResources service to make it easy to retrieve properties
- Integration with JavaMail to make sending email painless
- Integration with JavaMail and Velocity to allow you to send processed Velocity templates as email!
- Built using Ant
- Initial application development WAR archive creation tool using Ant
- Works cleanly with Servlet API 2.0 and higher
- Works with JDK 1.1.x and higher - 100% Pure Java



Turbine's Community

- Turbine has a mailing list with over 360 people on it
- There are over 30 people contributing code and documentation
- Used by people all over the world (many foreign contributors)
- Used as the basis for other large OSS projects like Jetspeed

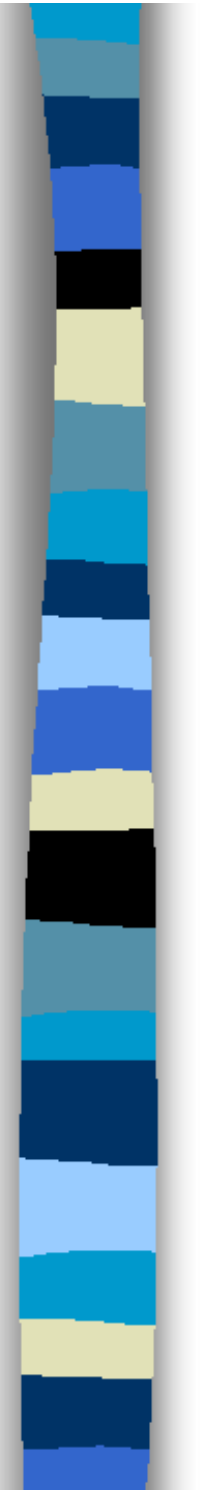


Turbine's Size

- Turbine now has almost 300 classes (and 5 slides of features!)
- Not all parts of Turbine have to be used
- Pick the parts you want and ignore the rest…until you find you need it. :-)

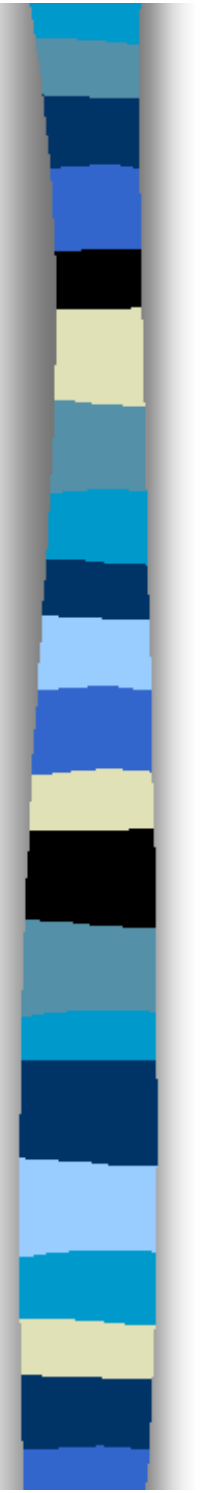
Template: HelloWorld.wm

```
$page.setTitle("Welcome to ApacheCon EU 2000")

This is an example WebMacro template!

<p>

Here is what was put into the context:

<p>

<b>$hello</b>
```



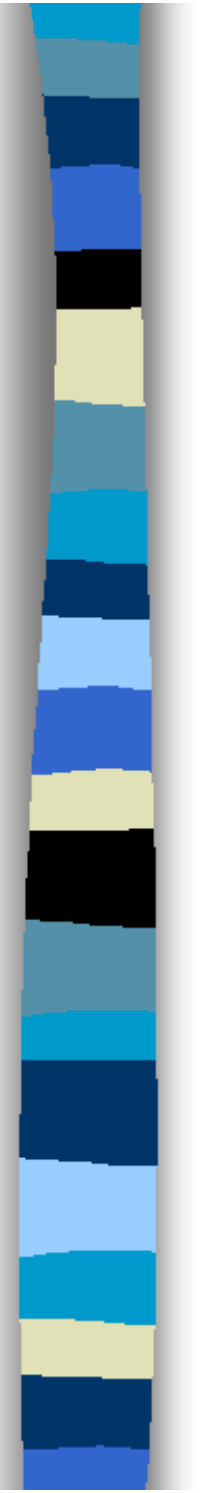
What is Model 2+1?

- Solves the problems of Model 2.
- Is not Model 3, but is instead Model 2 plus feature additions.
- Is a funny buzzword that I invented for this talk and has now caught on.



How is Turbine Model 2+1?

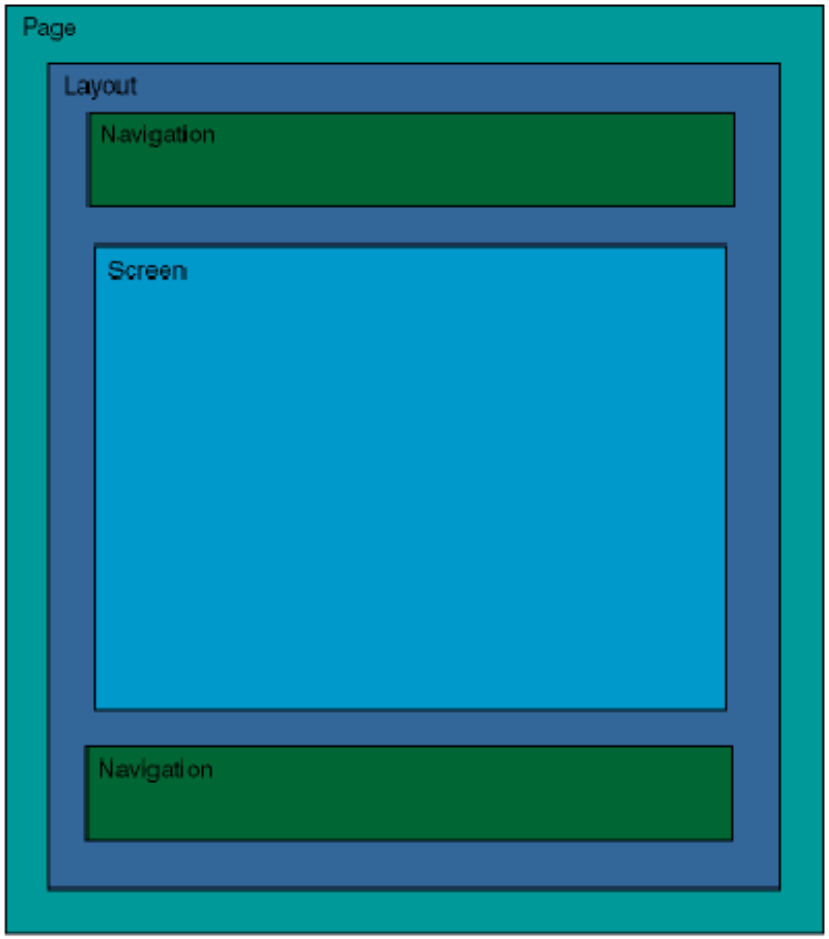
- Adds lots of reusable code
- Interfaces (and implementations) for Security, User Management and Services
- Turbine Modules for tying MVC together
  - Actions and ActionEvents
- Designed to be used as a webapp framework

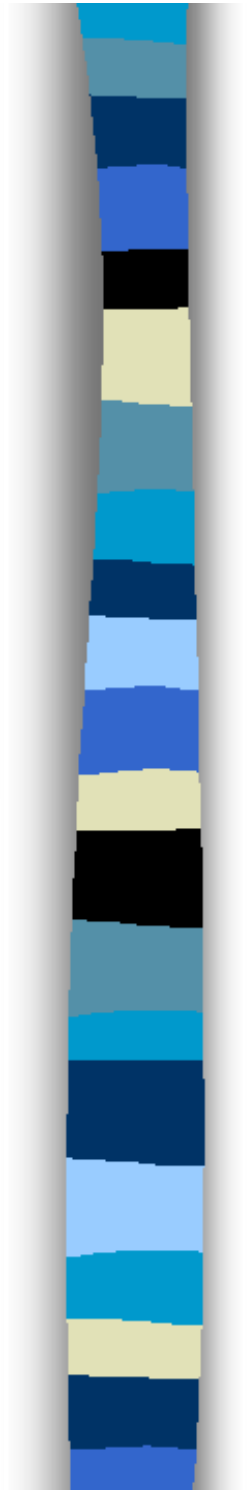


Turbine and MVC

- Turbine blurs the MVC model a bit
- Turbine provides the larger container for building MVC applications
- View is managed by one of the template solutions, but controlled by Turbine

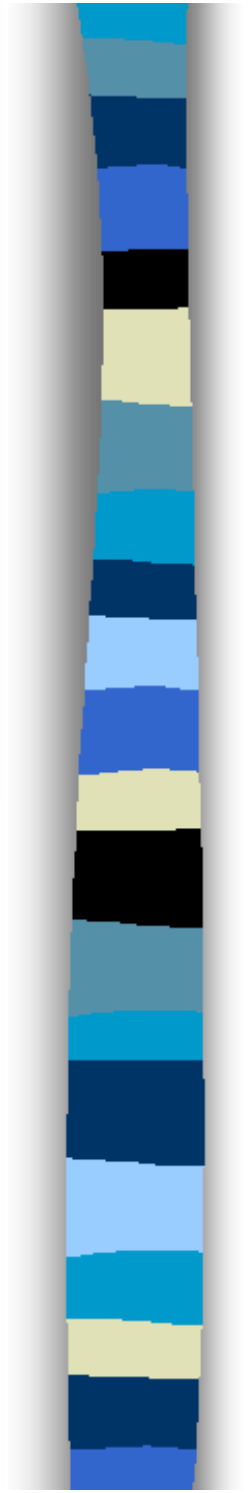
Picture of Turbine Modules





Java Directory Structure

- modules/
  - screens/
    - HelloWorld.java
  - actions/
    - MyAction.java
  - navigations/
    - No file needed right now.
  - layouts/
    - No file needed right now.



Template Directory Structure

- templates/
  - screens/
    - MyScreen.wm
  - actions/
    - MyAction.wm
  - navigations/
    - TopNav.wm and BottomNav.wm
  - layouts/
    - Default.wm

Screen: HelloWorld.java

```
public class HelloWorld extends WebMacroSiteScreen
{
    public void doBuildTemplate( RunData data, Context context )
        throws Exception
    {
        // the context object has already been setup for you!
        context.put ("hello", "this is a test...");
    }
}
```

Template: default.wm

```
$page.setStyleSheet("/myapp/style.css")
$page.setBgColor("#ffffff")
$navigation.setTemplate("/TopNav.wm")

<table width="100%">
    <tr>
        <td>$screen_placeholder</td>
    </tr>
   <tr>
<td>$navigation.setTemplate("/BottomNav.wm")</td>
   </tr>
</table>
```

Template: TopNav.wm

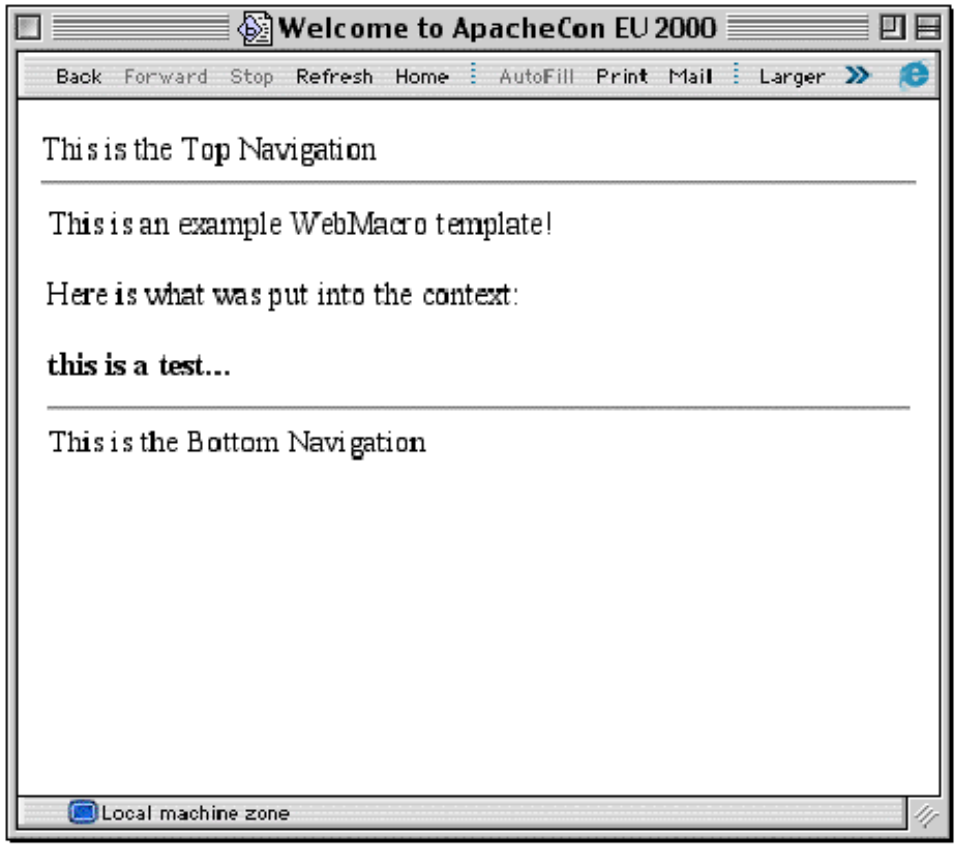
This is the Top Navigation
<hr>

Template: BottomNav.wm

<hr>

This is the Bottom Navigation

Example Output

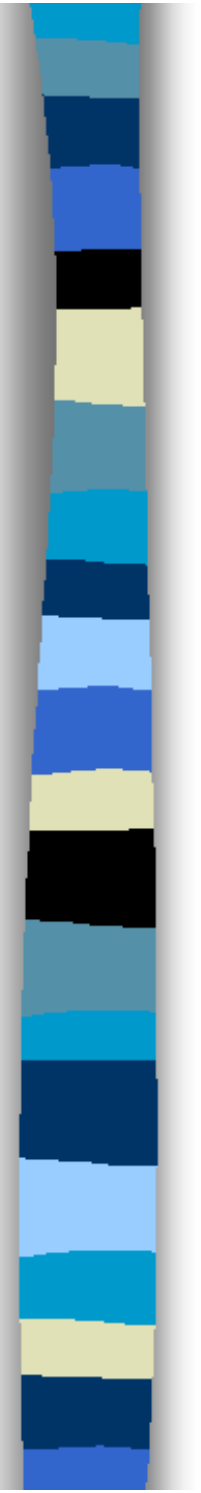


Actions - SampleScreen.wm

```
<form action="$link.setPage("Next.wm")" method=post>
<input type="hidden" name="action" value="MyAction">
<input type="text" name="f1" value="">
<input type="submit" name="eventSubmit_doSubmit"
value="Hit me!">
</form>
```

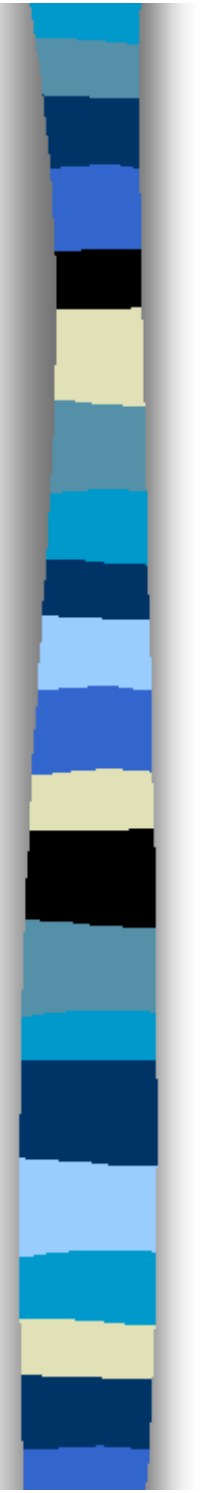
Actions - Java Code

```
public class MyAction extends WebMacroSiteAction
{
   public void doSubmit (RunData data, Context context)
      throws Exception
   {
       String  text = data.getParameters().getString(“f1”, “blank”);
       context.put ("username", text );
       data.setMessage(”Done!");
   }
   public void doPerform(RunData data, Context context)
      throws Exception
   {
      data.setMessage("Button not found!");
   }
}
```



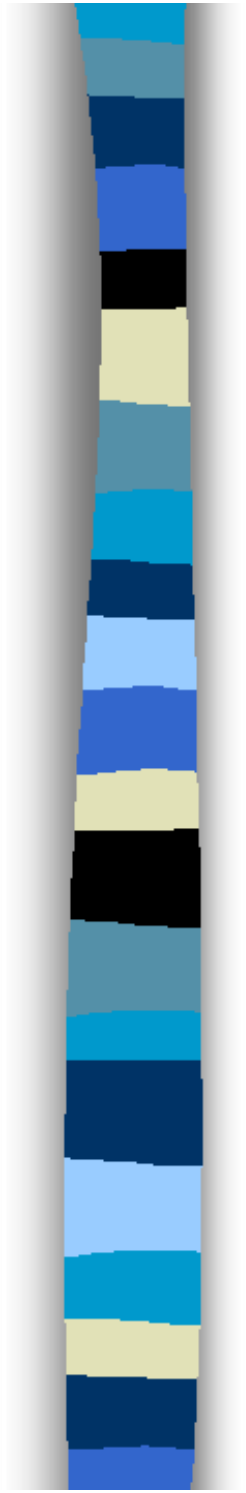
Turbine Developers Kit (TDK)

- Everything you need to start developing webapps with Turbine out of the box
- Everything is pre-configured. Just start Tomcat and go.
- Includes:
  - Tomcat
  - Turbine
  - Lots of .jars
  - Documentation



TDK Benefits

- Easily create a webapp with the directory structure and sample pages already in place
- All the external .jars are already there
- WebServer/Servlet engine
- Easily upgraded
- 100% Free



URL's

- http://java.apache.org/turbine/
- http://java.apache.org/turbine/tdk/
- http://jakarta.apache.org/velocity/
- http://www.webmacro.org/